BOLDx: R Package Reference Manual

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Prediction-assisted intervention, TITE, and combined dose-finding simulations.

Generated from the package Rd documentation; not a CRAN acceptance document.

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Source: https://github.com/wguo3773/BOLDx

Research use only; not for clinical dose assignment.

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bold_config

Configure a BOLD extension simulation

Description:

The prior at dose j is Beta(prior_mean[j] * prior_pess[j], (1 - prior_mean[j]) * prior_pess[j]). True untreated risks are separate simulation inputs and are never supplied to dose-assignment calculations.

Usage:

```
bold_config(
  target = 0.25,
  nmax = 21L,
  tau = 0.5,
  overdose_cutoff = 0.9,
  early_n = 12L,
  sensitivity = 0.9,
  specificity = 0.98,
  eligibility = 1,
  rho = 0.8,
  window = 3,
  accrual_rate = 2,
  arrival = c("uniform", "exponential"),
  timing = c("weibull", "uniform", "late"),
  prior_mean = NULL,
  prior_pess = 3,
  start_dose = 1L,
  pending_limit = 0.5,
  min_two = FALSE
)
```

Arguments:

target: Target untreated DLT probability, strictly between 0 and 5/7.

nmax: Patient cap, a positive multiple of three.

tau: BOLD decision-tail target in (0,1); not the DLT target.

overdose_cutoff: Posterior overdose elimination cutoff in (0,1).

early_n: Stop when this many patients have received the current dose and the next recommendation is unchanged. Use Inf to disable.

sensitivity, specificity: Prediction sensitivity and specificity for latent untreated DLT status, each in [0,1].

eligibility: Probability a patient is eligible for prediction.

rho: Probability an intervention prevents an otherwise-occurring DLT.

window: DLT assessment window, in months.

accrual_rate: Mean number of arrivals per month.

bold_config (continued)

arrival: Arrival-gap distribution, uniform or exponential.

timing: Generative conditional DLT-time distribution. The working follow-up model remains uniform for all choices.

prior_mean, prior_pess: Dose-specific vectors, or scalars recycled to all doses. NULL prior_mean uses target; PESS defaults to 3.

start_dose: Starting dose index, validated against each scenario.

pending_limit: Maximum pending fraction at the current dose before suspension; defaults to 0.5.

min_two: Require two completed outcomes before escalation.

Value:

A validated configuration list of class bold_config.

Examples:

```
cfg <- bold_config(nmax = 21, rho = 0.8)
cfg$target
```

simulate_bold

Simulate BOLD extensions and matched comparators

Description:

Fixed cohorts of three are used. Intervention failure is an observed DLT, not an extra latent branch. In combined mode, intervention follow-up resolves before the next dose decision. The implemented child-specific weighting rule is a sequential plug-in approximation, not exact Bayesian marginal evidence.

Usage:

```
simulate_bold(
  truth,
  method = c("prediction", "tite", "combined"),
  config = bold_config(),
  nsim = 100L,
  seed = 1L,
  mtd = NULL
)
```

Arguments:

truth: Nondecreasing vector of 2 to 10 untreated DLT probabilities, strictly between zero and one.

method: One or more of prediction, tite, combined, bold, boin, tite_boin.

config: A configuration from bold_config().

nsim: Number of replicate trials per method.

seed: Nonnegative integer seed. The caller's random state is restored.

mtd: True optimal dose index for evaluation. NULL chooses the dose closest to target (lower index breaks ties). Set 0 for all-too-toxic; K+1 means all doses are below target and the highest dose is optimal. This value affects summaries only, never dose decisions.

Value:

A bold_simulation list containing summary, trials, config, truth, mtd, and seed. Summary includes Monte Carlo standard errors.

Examples:

```
z <- simulate_bold(c(.10, .25, .45, .60),
  method = c("prediction", "bold", "boin"), nsim = 10, seed = 1)
z$summary
```

simulate_scenarios

Run a panel of dose-toxicity scenarios

Description:

Results are returned separately for each scenario, without averaging across MTD categories. Use explicit MTD labels for all-too-toxic and below-range cases.

Usage:

```
simulate_scenarios(
  scenarios,
  mtd = NULL,
  method = c("prediction", "tite", "combined"),
  config = bold_config(),
  nsim = 100,
  seed = 1
)
```

Arguments:

scenarios: Numeric matrix with one nondecreasing risk vector per row.

mtd: Optional integer vector giving each scenario's optimal dose.

method, config, nsim, seed: As in simulate_bold(). Seeds increment by scenario index; methods within a scenario share the same seed.

Value:

List with summary and trials data frames, each containing scenario.

Examples:

```
x <- rbind(c(.1,.25,.45,.6),c(.05,.1,.25,.45))
simulate_scenarios(x, mtd=c(2,3), method="prediction", nsim=5)$summary
```

beta_mixture

Inspect sequential weighted Beta components

Description:

Reproduces the simulation engine's child-specific plug-in weighting rule. It is order dependent and is not an exact Bayesian posterior update. Known outcomes should first be incorporated into a and b. An observed intervention failure is a known DLT: increment a rather than branching.

Usage:

```
beta_mixture(
  a,
  b,
  q = numeric(),
  labels = rep(-1L, length(q)),
  sensitivity = 0.9,
  specificity = 0.98,
  target = 0.25
)
```

Arguments:

a, b: Positive Beta parameters after resolved, nonlatent outcomes.

q: Remaining DLT-window fraction for each latent patient in [0,1]. Use 1 for an intervention patient without an observed DLT.

labels: Prediction labels: 1 positive, 0 negative, -1 uninformative.

sensitivity, specificity: Prediction operating characteristics.

target: Tail threshold for computing overdose probability.

Value:

List containing components, mean, and overdose_probability.

Examples:

```
beta_mixture(.75, 2.25, q = c(1, .5), labels = c(1L, -1L))
```